# Agentic AI in mathematical research

# A brief disclaimer

I am not an expert in this field. I have been exploring it out of curiosity and to stay up to date.

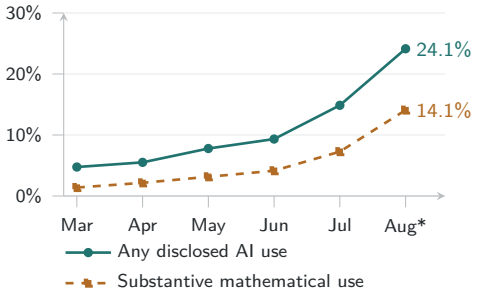
This talk neither advocates nor argues against using AI to write papers.

- A brief history and overview of mathematical research with LLMs, focusing on agentic approaches.
- What the literature tells us about successful approaches and their limitations.
- A case study where I build an AI research team to write a paper.

# AI use and mathematical output

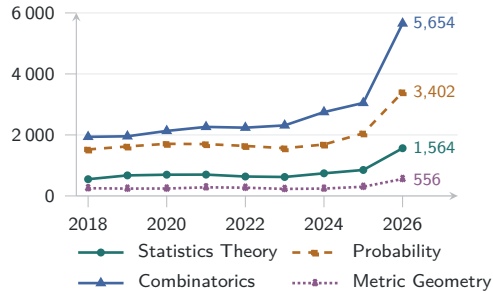
## AI disclosures on arXiv

Share of math-related submissions, 2026




## Submissions by field

New preprints, January–August each year

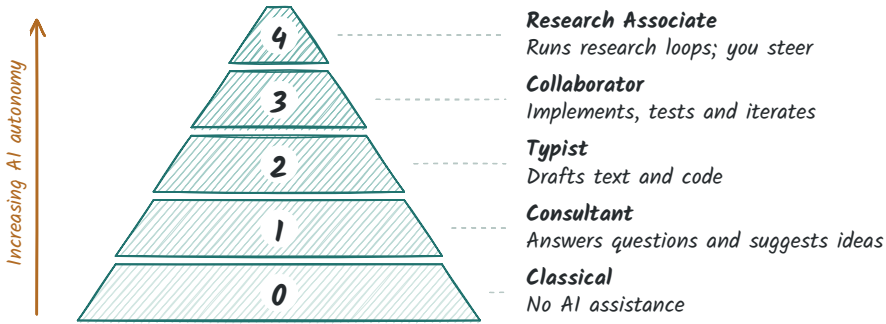



Left: 3,575 of 32,944 preprints disclosed AI use. *August covers 1–20.
Right: primary-category counts. Growth alone does not identify an AI effect.

Jin, Ke & Sui (2026), Table 1 **Explore arXiv Math Pulse** GitHub source.

# How have you used AI in research?

Which is the highest level you have used?




Adapted from Zimmer et al. (2026), *The Agentic Researcher*, Section 2.

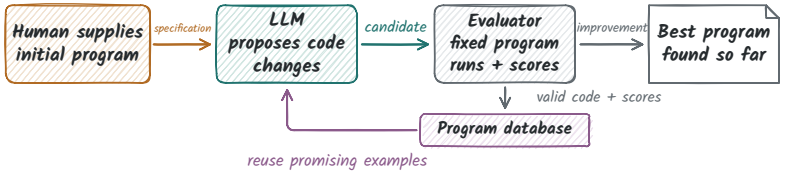
# A short history of math research with LLMs

## Agentic research systems over time

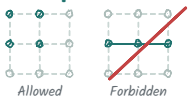
A brief history, with a focus on mathematics.

# FunSearch Romera-Paredes et al., *Nature* (2024)

An evolutionary approach with fixed evaluation criteria implemented in computer programs.




## Larger cap sets



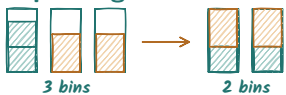

496 → 512

points in 8 dimensions

No three distinct points on a line in $\mathbb{F}_3^n$.

A new construction, not a proof of maximality.

## Online bin-packing heuristics



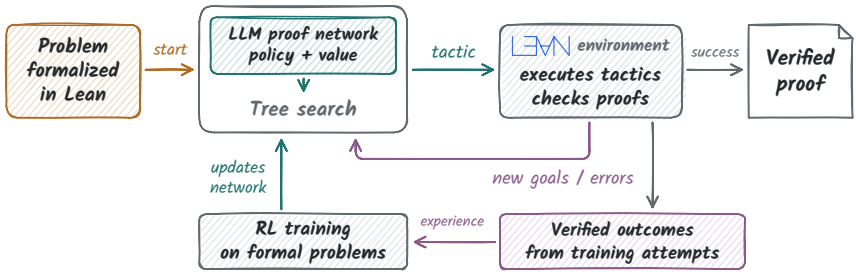

Use as few fixed-capacity bins as possible; place each item on arrival.

**AlphaEvolve (2025)** later extended this approach: $4 \times 4$ complex matrix multiplication with **48 scalar multiplications**, versus Strassen's 49.

Workflow: Fig. 1; results: Fig. 4 and Table 1. Romera-Paredes et al. (2024).

# AlphaProof Hubert et al., *Nature* (2026)

Another way to verify and evaluate mathematics is through formal verification in Lean.




**Search:** the LLM proposes tactics; Lean returns new proof states.

**Learning:** The model learns from proof attempts. For harder problems, it practises on similar ones.

**IMO 2024: silver-medal level — 28/42 points with AlphaGeometry 2.**

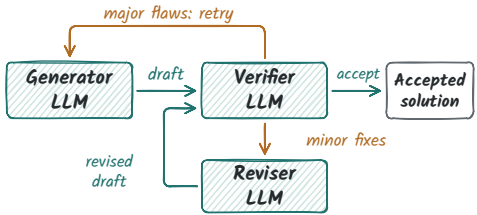
AlphaProof solved 3 problems; AlphaGeometry 2 solved 1.

Adapted from Figs. 1–2 and the AlphaProof section. *Nature* **651**, 607–613 (2026).

# Aletheia Feng et al., arXiv (2026)

We can also use an independent LLM for verification. This does not guarantee correctness.

## Gemini Deep Think agents




LLM judgment; no formal guarantee (Fig. 1).

## Reported mathematical results

- Erdős: 2 full + 2 partial new solutions from 700 problems, assessed by experts.

And other mathematical achievements.

## Other LLM research systems

**The AI Scientist** *Nature* (2026)
Ran ML experiments and wrote a paper that passed a workshop's first review round.

**Google Co-Scientist** *Nature* (2026)
Agents proposed and refined leukaemia drug combinations, validated in laboratory experiments.

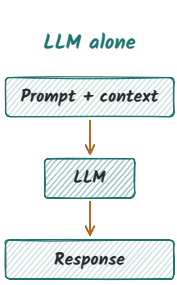
**Virtual Lab** *Nature* (2025)
LLM agents designed SARS-CoV-2 nanobodies with binding confirmed in laboratory tests.

**OpenAI: Navier–Stokes** report (2026)
LLM teams explored different approaches; the successful team ran about **10,000 agents** concurrently. **Human researchers** redirected effort and used Codex to share insights between teams.

# Agentic systems

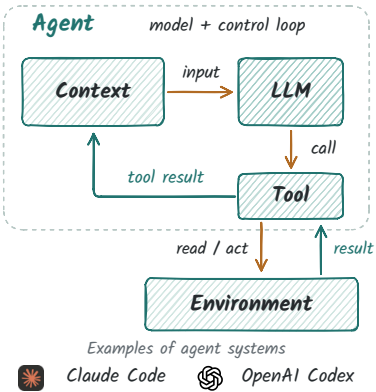
## From LLMs to research teams

# Environment, context, tools and agents




$$Y = f_\theta(p, c; \xi)$$

$p$: prompt; $c$: context
$\theta$: fixed model
$\xi$: randomness




**Environment**
The workspace an agent can access: files, data, code and external services.

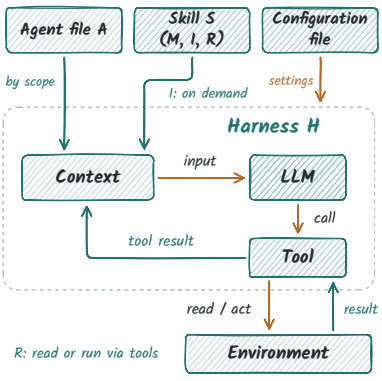
**Context**
The information given to the LLM now:
instructions, messages and tool results.

**Tool**
An interface to observe or change the environment, e.g. search or Python.

**Agent**
An LLM-based system that uses tools and chooses its next step from results.

Based on OpenAI's Codex harness documentation and Claude Code's agentic loop.

# Skills, agent files and the harness




**Agent instruction file** $A = (D, I_A)$
$D$: scope; $I_A$: standing instructions.
E.g. `AGENTS.md`: project guidance loaded into the context within its scope.

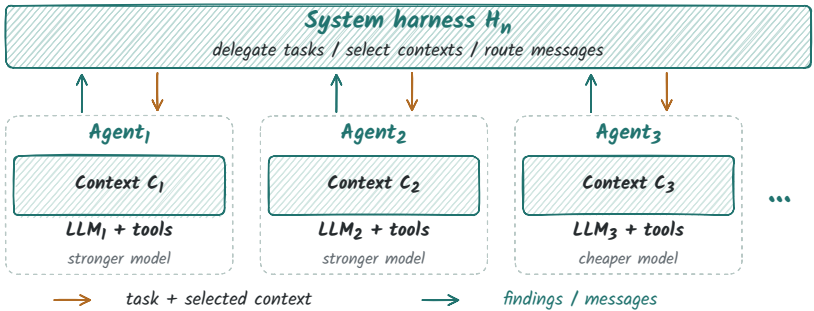
**Skill** $S = (M, I, R)$
$M$ — **metadata**: name, what the skill does, and when to use it.
$I$ — **instructions**: method, decision rules, checks and expected output.
$R$ — **resources**: optional documents, code, examples and templates.

**Harness** $H$
Orchestration software. A configuration file sets roles, models, tools, skills and routing.

Working abstractions for this talk. Implementation: skills, `AGENTS.md`, Codex harness (OpenAI Docs).
Foundations: ReAct — Yao et al. (2022), Voyager — Wang et al. (2023), SWE-agent — Yang et al. (2024).

# Multi-agent systems: who sees what?

Recall $Y = f_\theta(p, c; \xi)$. So context engineering is very important.




$H_n$ is the software coordinating $n$ agents, their contexts and communication.

A configuration file sets roles/models/tools; skills and agent files supply instructions.

Same or different models. Sources: AutoGen (Wu et al., 2023); OpenAI agent configuration.

# Why manage context and communication?

Very simply, an LLM produces random text $f(\text{context})$, whose distribution **depends strongly on the context**.

**1. Keep context focused** Liu et al. (2024); Modarressi et al. (2025)
**Problems:** context distraction and context clash.
Give each agent the information it needs; long histories can obscure relevant details.

**2. Give review a fresh context** Sharma et al. (2023); Dhuliawala et al. (2023)
**Problems:** context poisoning and context confusion.
Review the draft and criteria without the writer's history; verify independently.

**3. Match the model to the task** Ong et al., *RouteLLM* (2024)
Routing between stronger and cheaper models can improve the balance of cost and quality.

**4. Coordinate selectively** Du et al. (2023); Wang et al. (2025)
Let agents share useful findings. Removing agents or messages that repeat the same work can save time and improve results.

Effects depend on the task and models. Fresh-context review is a design inference from bias and verification studies.

# Heuristics for agentic research

| Heuristic | Intuition |
|---|---|
| **Ask for a proof or a counterexample.** [1] | Let the agent show that the claim is false; do not let it silently change the problem to obtain a proof. |
| **Delegate precise mathematical obligations.** [2] | Explicit assumptions and dependencies make partial progress and remaining gaps visible. |
| **Explore distinct approaches independently.** [3] | Different starting points can reveal approaches that others miss. |
| **Preserve each claim's status and evidence.** [4] | Reuse must not turn a conjecture or rejected argument into an "established lemma". |
| **Fix identified errors; stop if no progress is made.** [5] | Check that each error was repaired. Repeated approval is not evidence of a correct proof. |

Evidence-informed design heuristics: support ranges from benchmark experiments to research-system case studies.

[1] Petrov et al. (2025); [2] Zhao et al. (2026); [3] Zhu et al. (2026);
[4] Xu et al. (2026); [5] Zheng et al. (2026).

# Agentic research: a worked example

## What do we need?

- An **agent system**: in my case, Codex.
- A precise **mathematical problem statement**.
- Relevant **related work**: definitions, known results and useful methods.
- `instruction.md`: the task, constraints and expected output.
- `skills.md`: an index of reusable methods and checks.
- `AGENTS.md`: standing project instructions for the agents.

Here, `instruction.md` and `skills.md` are our project documents.
Each actual Codex skill has its own `SKILL.md`.

# Step 1: the precise problem

A **simple point process** $X$ on $\mathbb{R}^d$ is a random set of distinct points, finite in bounded regions. Its **count** in $B$ is

$$X(B) = \#(X \cap B).$$

The **factorial moment measure** counts distinct, ordered $n$-tuples:

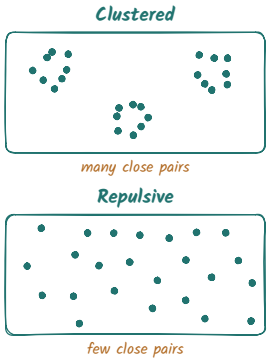
$$\alpha^{(n)}(B_1 \times \cdots \times B_n) = \mathbb{E} \sum_{(x_1,\ldots,x_n) \in X^n_{\neq}} \prod_{i=1}^{n} \mathbf{1}_{B_i}(x_i).$$

At order two,

$$\alpha^{(2)}(B_1 \times B_2) = \mathbb{E}[X(B_1)X(B_2)] - \mathbb{E}[X(B_1 \cap B_2)].$$

**Interpretation:** the expected number of ordered pairs of distinct points, with the first in $B_1$ and the second in $B_2$.
A point is never paired with itself.

Waagepetersen (2022), slides 7–13; Last and Penrose, Section 4.4.




Schematic point patterns.

# Step 1: the precise problem

When $\alpha^{(n)}$ has a locally integrable density, the **product density** is defined by

$$\alpha^{(n)}(A) = \int_A \rho^{(n)}(x_1, \ldots, x_n)\, \mathrm{d}x_1 \cdots \mathrm{d}x_n.$$

At first order, $\rho^{(1)}$ is the density (derivative) of the **mean count** $\alpha^{(1)}(B) = \mathbb{E}[X(B)]$:

$$\rho^{(1)}(x) = \frac{\mathrm{d}\alpha^{(1)}}{\mathrm{d}x}(x), \qquad \rho^{(2)}(x, y) = \frac{\mathbb{E}[X(\mathrm{d}x)X(\mathrm{d}y)]}{\mathrm{d}x\, \mathrm{d}y} \quad (x \neq y).$$

With positive intensity densities $\rho^{(1)}(x_i)$, define the ***n*-point correlation function**:

$$g_n(x_1, \ldots, x_n) = \frac{\rho^{(n)}(x_1, \ldots, x_n)}{\prod_{i=1}^n \rho^{(1)}(x_i)}.$$

**Stationarity:** shifting all points leaves their joint law unchanged: $X + t \overset{d}{=} X$ for every $t \in \mathbb{R}^d$. Then $\rho^{(1)}(x) = \lambda$ is constant and $g_n = \rho^{(n)}/\lambda^n$ (assume $0 < \lambda < \infty$).

Relative to Poisson with the same intensity:

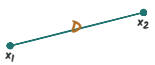
$g_n > 1$**: more such tuples.** $g_n < 1$**: fewer such tuples.**

For pairs: clustering or repulsion at the specified locations and scale. $g_2 = 1$ alone does not imply Poisson.

Waagepetersen (2022), slides 9–13; Last and Penrose, Section 4.4.

# Step 1: the precise problem

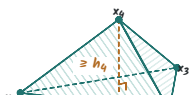
Fix $M, \lambda_0, h_3, h_4 > 0$.
Find **one stationary, simple** point process $X$ on $\mathbb{R}^3$ with intensity $\lambda_0$, finite local moments through order four and product densities.



Distance D



Area A



Volume V

For some constants $C_2, C_3, C_4 > 0$, require **simultaneously**:

$$g_2(x_1, x_2) = \frac{C_2}{D}, \qquad g_3(x_1, x_2, x_3) = \frac{C_3}{A}, \qquad g_4(x_1, x_2, x_3, x_4) = \frac{C_4}{V}.$$

Almost everywhere on nondegenerate configurations of diameter $\leq M$, with every triangle altitude $\geq h_3$ and every tetrahedron altitude $\geq h_4$.

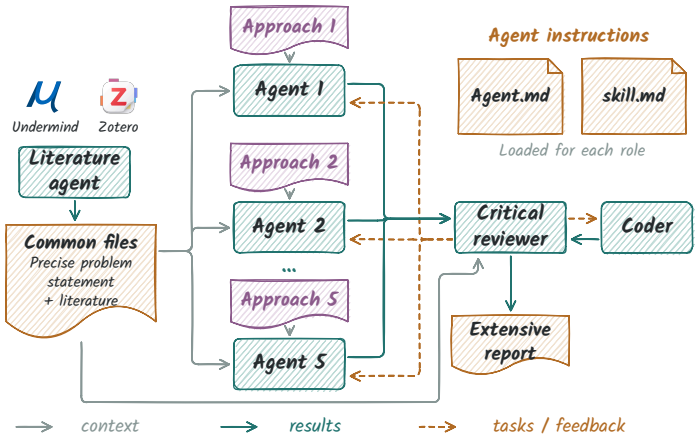
$C_2, C_3, C_4$ are chosen jointly with the process, not prescribed independently.

Current construction: sufficiently small $\lambda_0$; arbitrary prescribed intensity remains open.

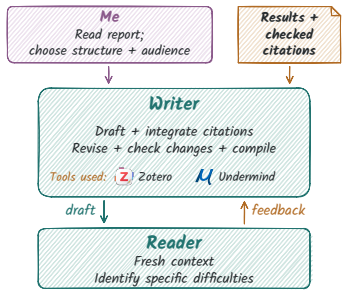
# Step 2: configuration/agent files and execution

**Research approaches**

1. Poisson cluster process
2. Cox process
3. Hawkes process
4. An approach of the agent's choice
5. Prove non-existence



# Step 3: Writing the paper




**Complete-task statistics**

| Measure | Complete task |
|---|---|
| Input tokens | 131,866,681 |
| Cached input (included above) | 128,329,472 |
| Uncached input | 3,537,209 |
| Output (includes reasoning) | 666,705 |
| Reasoning (subset of output) | 286,459 |
| **Total tokens** | **132,533,386** |
| Task-turn runtime | **3:47:06** |
| Research coordinator runtime | **3:36:43** |
| API-equivalent cost | **$78.81** |

**Writing stage: token use and cost**

| Participant | Uncached input | Cached input | Output/reasoning | Cost |
|---|---|---|---|---|
| Coordinating writer | 0.30M | 3.5M | 0.15M | $5.60 |
| Simulation specialist | 0.15M | 1.4M | 0.10M | $3.16 |
| Fresh reader | 0.06M | 0.1M | 0.02M | $0.68 |
| **Total** | **0.51M** | **5.0M** | **0.27M** | **$9.44** |

# Results and personal feedback

## Poisson cluster model

Homogeneous Poisson parents $\Pi$ each generate **0–3 children**, with jointly chosen offsets $C_z$:

$$X = \bigcup_{z \in \Pi} \big( \{z\} \cup (z + C_z) \big).$$

$C_z$ contains only child offsets. **The parent is retained**, giving 1–4 points per cluster.

**How many points?** The solution gives a probability distribution over cluster sizes $j = 1, \ldots, 4$.

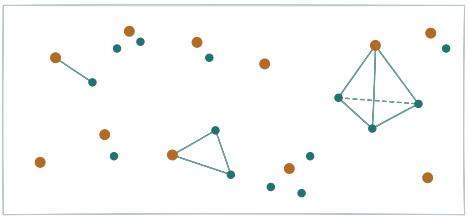
**Where are they placed?** For $j \geq 2$, the normalised density

$$\frac{F_j(z, x_1, \ldots, x_{j-1})}{Z_j}$$

describes how the $j - 1$ children are jointly distributed around the parent $z$.

## Personal feedback

| Aspect | Feedback |
|---|---|
| Mathematical correctness | **Positive** |
| Research and process documentation | **Positive** |
| Text writing | **Medium** |
| Paper structuring | **Negative** |
| Mathematical explanation | **Medium** |



# Future of agentic research

## What I expect to change

Stronger models can increasingly plan, divide work into tasks, use tools and delegate to subagents.

For a precise mathematical problem, **one initial prompt may increasingly suffice**. This can still mean substantial autonomous work, with no guarantee of success.

As platforms provide more harness functions, we will have **less to build ourselves**.

I also expect explanations to become **less sloppy and more useful for learning**.

## Where a custom harness still helps

- Keep the exact target, hypotheses and criteria for success fixed.
- Require separate review and formal verification.
- Keep human checkpoints for direction and interpretation.
- Record results, failed approaches, sources and dependencies.
- Set compute budgets, stopping rules and model choice.
- When writing papers with AI, set your preferences for writing and illustration styles.

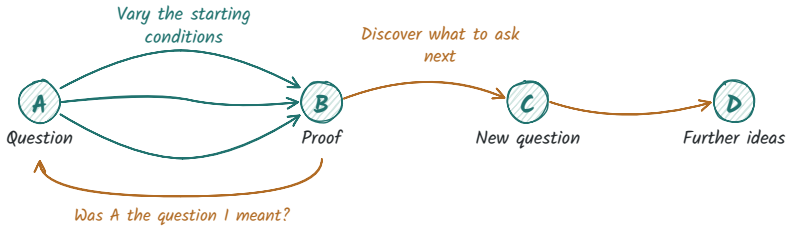
**A reported example.** For its October release, OpenAI says almost all results came from one prompt to one agent. Some may have needed retries; the full workflow has not been independently reproduced.

Scientific American, October 2026; OpenAI release details.

# Personal take

I find this exciting: an elegant solution, with reasoning that feels human.
Still, something is missing when I do not discover the ideas myself.

**Faster proofs let me explore more. Research was never a straight line.**




**Reaching B only helps me if I understand the mathematics.**

**My concern as a PhD student:** outsourcing too much may bypass the learning.
A PhD is still an education. Producing results and developing understanding can come apart.

Related: *A Severe Misalignment of AI in Mathematics* (joint declaration, 11 September 2026).

# Thank you!

## Questions and discussion

Slides and, later, the experiments:

github.com/MathisRost/agentic-ai-math

